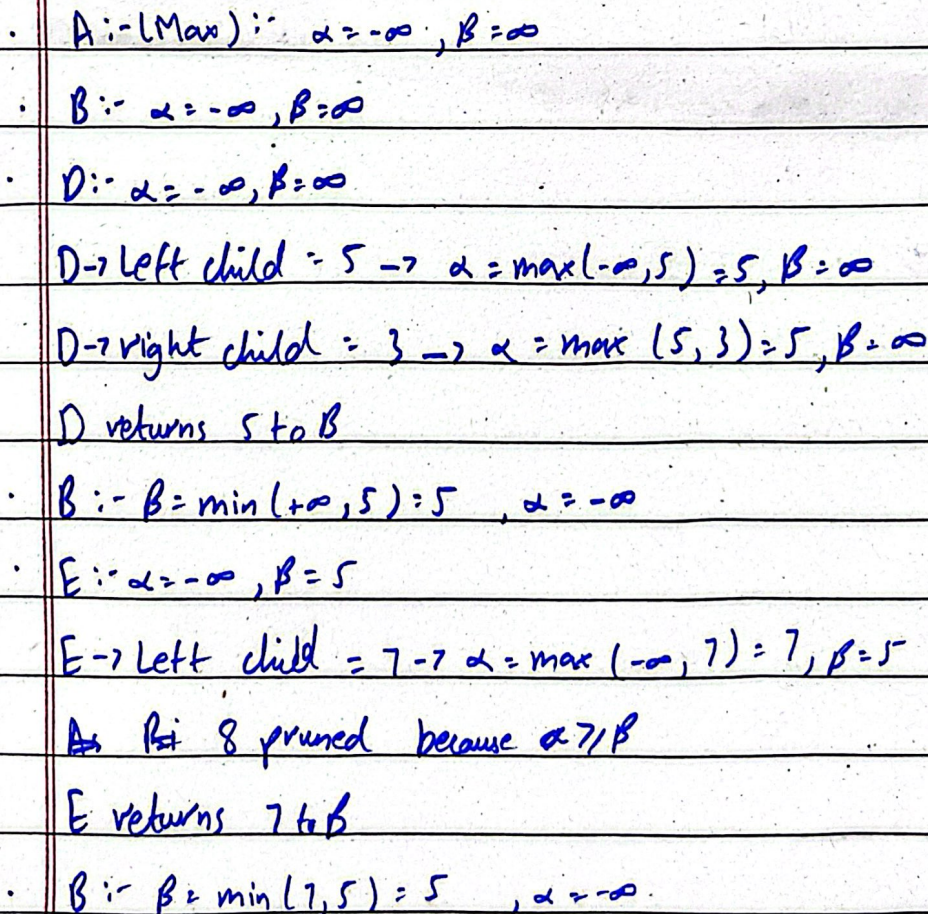


AI Project 1 Part 2



(2)

Maha Faisal

BSc 24048

Date: _____

$$\cdot A: \alpha = 5, \beta = \infty$$

$$\cdot C: \alpha = 5, \beta = \infty$$

$$\cdot F: \alpha = 5, \beta = \infty$$

$$F \rightarrow \text{Left child} = 3 \rightarrow \alpha = \max(5, 3) = 5, \beta = \infty$$

$$F \rightarrow \text{right child} = 6 \rightarrow \alpha = \max(5, 6) = 6, \beta = \infty$$

$$F \text{ returns } 6 \text{ to } C; \alpha = 6, \beta = \infty$$

$$\cdot C: \beta = \min(\infty, 6) = 6$$

$$\text{Hence: } \alpha = 5, \beta = 6$$

$$\cdot G: \alpha = 5, \beta = 6$$

$$G \rightarrow \text{Left child} = 9 \rightarrow \alpha = \max(5, 9) = 9, \beta = 6$$

$\hookrightarrow$ pruned because $\alpha > \beta$ (9 > 6)

$$G \text{ returns } 9 \text{ to } C; \alpha = 9, \beta = 6$$

~~pruned because $\alpha > \beta$ (9 > 6)~~

$$\cdot C: \alpha = 5, \beta = 6$$

$$\cdot A: \alpha = \max(5, 6) = 6$$

$$\beta = \infty$$